

INTELLIQUERY: AI-POWERED CONVERSATIONAL BUSINESS INTELLIGENCE DASHBOARD PLATFORM

Software Requirements Specification

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Abstract

IntelliQuery is an AI-powered conversational dashboard platform that revolutionizes business intelligence accessibility through natural language interaction. The system integrates with Power BI data sources but presents insights through a custom-built intelligent dashboard where users can ask questions in plain English and receive dynamic visualizations, reports, and actionable insights. This Software Requirements Specification document provides a comprehensive overview of the functional and non-functional requirements, use cases, user interface specifications, and technical considerations for the IntelliQuery platform. The system employs advanced Natural Language Processing, contextual memory management, and agentic AI architecture to eliminate technical barriers in data analysis while ensuring enterprise-grade security and scalability. Built in collaboration with Folio3 Software Solutions, IntelliQuery addresses the critical need for democratizing business intelligence access in Pakistani enterprises and beyond.

Keywords: Conversational AI, Business Intelligence, Custom Dashboard, Natural Language Processing, Data Visualization, Enterprise Analytics, Software Requirements, Use Cases, Agentic AI, LangGraph

Chapter 1

Problem Statement

1.1 Introduction to the Problem Domain

Business intelligence tools have become essential for modern organizations seeking to leverage data for strategic decision-making. However, these powerful platforms create significant accessibility barriers that prevent non-technical stakeholders from effectively utilizing organizational data. Research indicates that 85% of business users struggle with traditional BI tools that require knowledge of SQL, DAX, or complex query languages. Furthermore, analysts spend 60-80% of their time on data preparation and basic query creation rather than actual analysis, representing a massive inefficiency in organizational resources.

1.2 Current State of Business Intelligence Access

The current landscape of business intelligence presents several critical challenges that limit its effectiveness and adoption across organizations:

1.2.1 Technical Complexity Barriers

Traditional BI platforms like Power BI, Tableau, and Qlik require users to understand complex concepts including data models, relationships, measures, calculated columns, and query languages. Non-technical stakeholders—including executives, managers, and business analysts—depend entirely on IT teams or specialized BI analysts to generate reports and answer data questions. This dependency creates bottlenecks, delays decision-making, and limits the democratization of data access that organizations strive to achieve.

1.2.2 Time and Resource Inefficiency

Manual report generation is time-consuming and creates significant operational bottlenecks. When business users need insights, they must submit requests to technical teams, wait for

queries to be written and executed, and often go through multiple iterations to refine results. This process can take hours or even days, by which time the business context may have changed. The cumulative effect is that organizations make decisions with outdated information or avoid data-driven decision-making altogether due to the friction involved.

1.2.3 Static Dashboard Limitations

Existing dashboards provide fixed views of data that cannot adapt to dynamic business questions. While dashboards can display predefined metrics and visualizations, they fail when users have follow-up questions, need different perspectives, or want to explore data in ways not anticipated by the dashboard designer. Users must either accept the limited insights provided or submit new requests for custom reports, perpetuating the cycle of dependency and delay.

1.2.4 Context Loss in Analysis

When navigating between different BI tools, reports, and dashboards, users frequently lose the analytical thread of their investigation. Each new view or report requires starting from scratch, remembering previous findings, and manually connecting insights across different sources. This fragmentation makes it difficult to conduct coherent, multi-step analysis and often results in incomplete or superficial insights.

1.2.5 Pakistani Enterprise Context

Pakistani enterprises face additional challenges that compound these universal BI accessibility issues:

- **Limited BI Expertise:** There is a scarcity of trained BI professionals who can effectively implement and manage business intelligence systems, making organizations heavily dependent on external consultants or struggling with underutilized BI investments.
- **High Consultant Costs:** Specialized BI consultants command premium rates that many Pakistani enterprises find prohibitive, especially for ongoing support and maintenance needs.
- **Language and Terminology Barriers:** International BI platforms often lack support for local business terminology and practices, creating additional friction for users who must mentally translate between standard BI concepts and their specific business context.
- **Infrastructure Limitations:** Many organizations lack the robust IT infrastructure assumed by enterprise BI platforms, requiring solutions that can operate effectively with limited resources.

- **Cultural Resistance to Technology Adoption:** Organizations may resist adopting new technologies due to skepticism about reliability, preference for traditional decision-making methods, or concerns about displacing existing staff.

1.3 Project Scope

IntelliQuery will focus on delivering a comprehensive conversational business intelligence platform with the following scope:

1.3.1 Core Functionality

1. **Natural Language Query Processing:** Convert business questions expressed in plain English into structured Power BI queries and return meaningful results.
2. **Custom Conversational Dashboard:** Provide a web-based interface where users interact with their data through chat and receive dynamically generated visualizations.
3. **Contextual Memory System:** Maintain conversation history and context to enable multi-turn analytical conversations.
4. **Power BI Integration:** Seamlessly connect to existing Power BI data sources while leveraging enterprise data models and security protocols.
5. **Dynamic Visualization Generation:** Automatically create appropriate charts and visualizations based on query results and data characteristics.

1.3.2 User Roles

The system will support three primary user roles:

- **Business Users:** End-users who query data and receive insights through natural language conversation.
- **Administrators:** Users who manage system configuration, user access, and data source connections.
- **System Administrators:** Technical users responsible for system maintenance, monitoring, and troubleshooting.

1.3.3 Out of Scope

The following capabilities are explicitly out of scope for the current implementation:

- Direct modification of Power BI reports and dashboards

- Data warehouse or ETL functionality
- Mobile native applications (web-responsive design will be provided)
- Voice-based query input (planned for future releases)
- Multi-language support beyond English (planned for future releases)

Chapter 2

System Requirements

2.1 List of Functional Requirements

ID	Requirement	Description
FR1	Natural Language Query Processing	System shall accept business questions in plain English and convert them into structured queries against Power BI data sources.
FR2	Intent Classification	System shall identify user intent from natural language input (e.g., data retrieval, comparison, trend analysis, drill-down).
FR3	Entity Extraction	System shall extract relevant business entities from queries (dates, metrics, dimensions, filters).
FR4	Contextual Memory Management	System shall maintain conversation history and context across multiple user interactions within a session.
FR5	Dynamic Visualization Generation	System shall automatically select and generate appropriate chart types based on data characteristics and query intent.
FR6	User Authentication	System shall implement secure user authentication with support for Azure Active Directory integration.
FR7	Chat Interface	System shall provide a modern, responsive chat interface for user interactions.
FR8	Visualization Export	System shall allow users to export generated visualizations and insights in multiple formats (PDF, Excel, PowerPoint).
FR9	Query History	System shall maintain a searchable history of user queries and generated insights.
FR10	Follow-up Questions	System shall support follow-up questions that build upon previous queries without requiring users to repeat context.

ID	Requirement	Description
FR11	Clarification Dialogue	System shall request clarification from users when queries are ambiguous or insufficient information is provided.
FR12	Error Handling	System shall provide clear, actionable error messages when queries cannot be processed successfully.
FR13	Real-time Dashboard Updates	System shall update visualizations in real-time as conversation progresses.
FR14	Query Suggestions	System shall provide context-aware suggestions for related queries users might want to explore.
FR15	Confidence Scoring	System shall provide confidence scores for query interpretation to indicate reliability of results.
FR16	Multi-Dataset Support	System shall support querying across multiple Power BI datasets within a single conversation.
FR17	Data Source Configuration	System administrators shall be able to configure connections to Power BI workspaces and datasets.
FR18	User Management	Administrators shall be able to create, modify, and deactivate user accounts with appropriate role assignments.

Table 2.1: Functional Requirements

2.2 List of Non-Functional Requirements

ID	Requirement	Description
NFR1	Response Time	System shall process simple queries and return results within 3 seconds under normal load conditions.
NFR2	Complex Query Performance	System shall process complex queries requiring multiple data sources within 10 seconds.
NFR3	System Availability	System shall maintain 99% uptime during business hours (8 AM - 8 PM local time).
NFR4	Concurrent Users	System shall support at least 50 concurrent users without performance degradation.
NFR5	Scalability	System architecture shall be designed to scale horizontally to support increasing user loads.
NFR6	Data Security	All sensitive data shall be encrypted in transit (TLS 1.3) and at rest (AES-256).
NFR7	Authentication Security	System shall implement industry-standard authentication protocols including OAuth 2.0 and JWT tokens.

ID	Requirement	Description
NFR8	Browser Compatibility	System shall function correctly on latest versions of Chrome, Firefox, Safari, and Edge browsers.
NFR9	Mobile Responsiveness	User interface shall be responsive and functional on tablets and mobile devices.
NFR10	Accessibility	System shall meet WCAG 2.1 Level AA accessibility standards.
NFR11	Usability	Non-technical users shall be able to use core features with less than 15 minutes of training.
NFR12	Query Accuracy	System shall achieve greater than 90% accuracy in intent classification and entity extraction.
NFR13	API Rate Limiting	System shall implement rate limiting to prevent API abuse and ensure fair resource allocation.
NFR14	Error Recovery	System shall gracefully handle and recover from errors without requiring system restart.
NFR15	Audit Logging	System shall maintain comprehensive audit logs of all user actions and system events.
NFR16	Backup and Recovery	System shall support automated backups with recovery time objective (RTO) of 4 hours.
NFR17	Maintainability	System code shall follow industry best practices and be well-documented to facilitate maintenance.
NFR18	Integration	System shall provide RESTful APIs for potential integration with other enterprise systems.
NFR19	Monitoring	System shall provide real-time monitoring dashboards for performance and health metrics.

Table 2.2: Non-Functional Requirements

Chapter 3

User Interface Requirements

3.1 Business User Portal

3.1.1 Dashboard (Home Page)

The business user dashboard serves as the primary entry point for users to interact with their data through conversation.

Components:

- **Chat Interface:** Left panel featuring a modern chat interface where users can type natural language queries
- **Visualization Panel:** Right panel displaying dynamically generated charts and insights based on conversation
- **Recent Conversations:** Sidebar showing recent conversation threads for quick access
- **Quick Insights:** Top section displaying key metrics and suggested questions based on user's role and data access

Chat Interface Features:

- Message history with timestamps
- Query suggestions based on context
- Ability to edit and resubmit queries
- Clear visual distinction between user messages and AI responses
- Inline visualization previews within chat

Visualization Panel Features:

- Full-screen chart view
- Interactive charts with hover details and cross-filtering
- Export options (PDF, Excel, PowerPoint, PNG)
- Share functionality for distributing insights to team members
- Zoom and pan capabilities for detailed exploration
- Side-by-side comparison mode for multiple visualizations

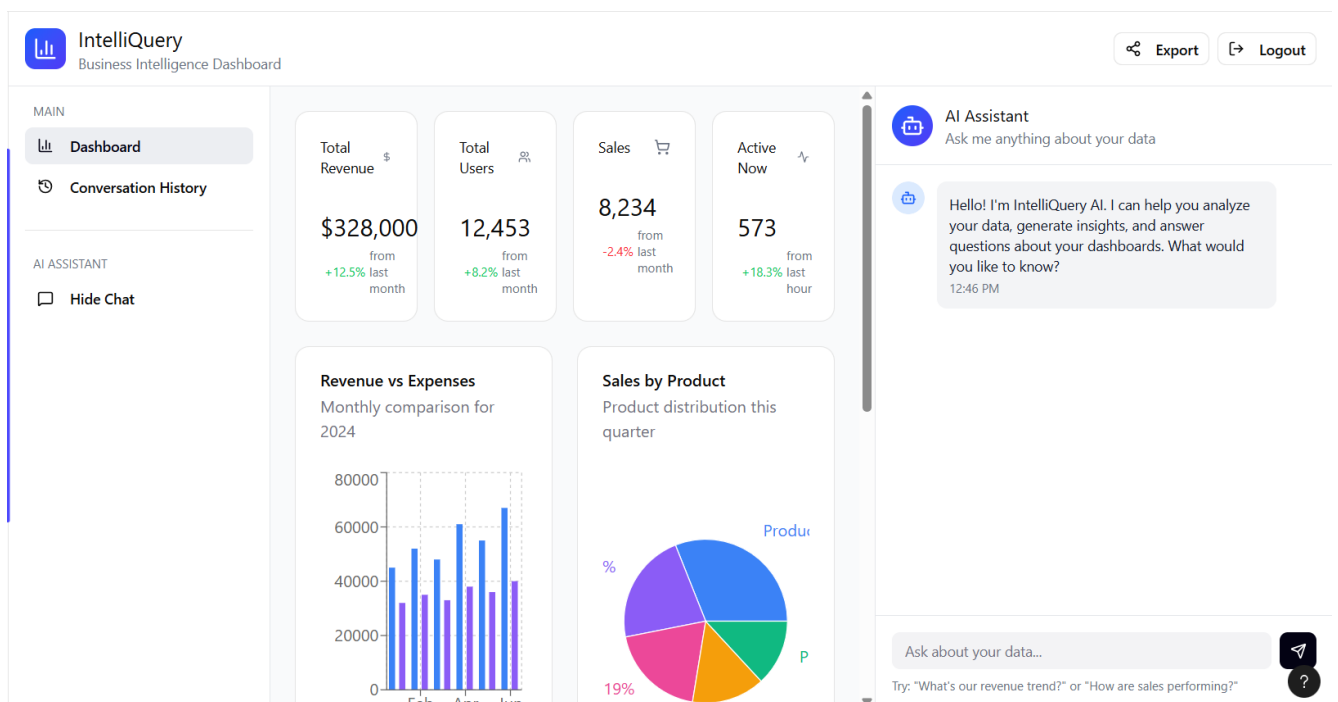


Figure 3.1: Business User Dashboard mockup showing split-screen layout with chat on left and visualizations on right

3.1.2 Conversation History Page

- **Search Functionality:** Search through past conversations by keywords, dates, or topics
- **Conversation Cards:** Each conversation displayed as a card showing:
 - Conversation title (automatically generated from first query)
 - Date and time of conversation
 - Preview of insights generated
 - Quick action buttons (resume, export, share, delete)

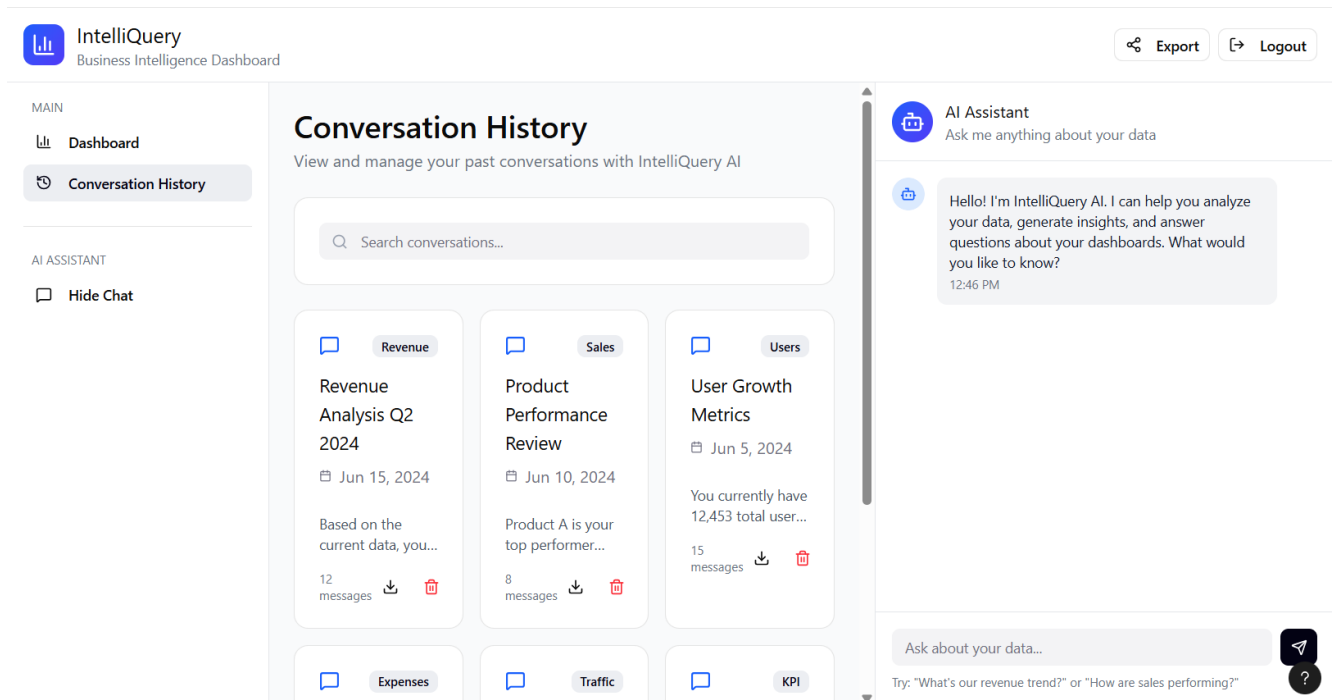


Figure 3.2: Conversation History page mockup

3.1.3 Export and Share Page

- **Export Formats:** Select from PDF report, Excel workbook, or PNG images
- **Sharing Options:** Share via email, download, or generate shareable links

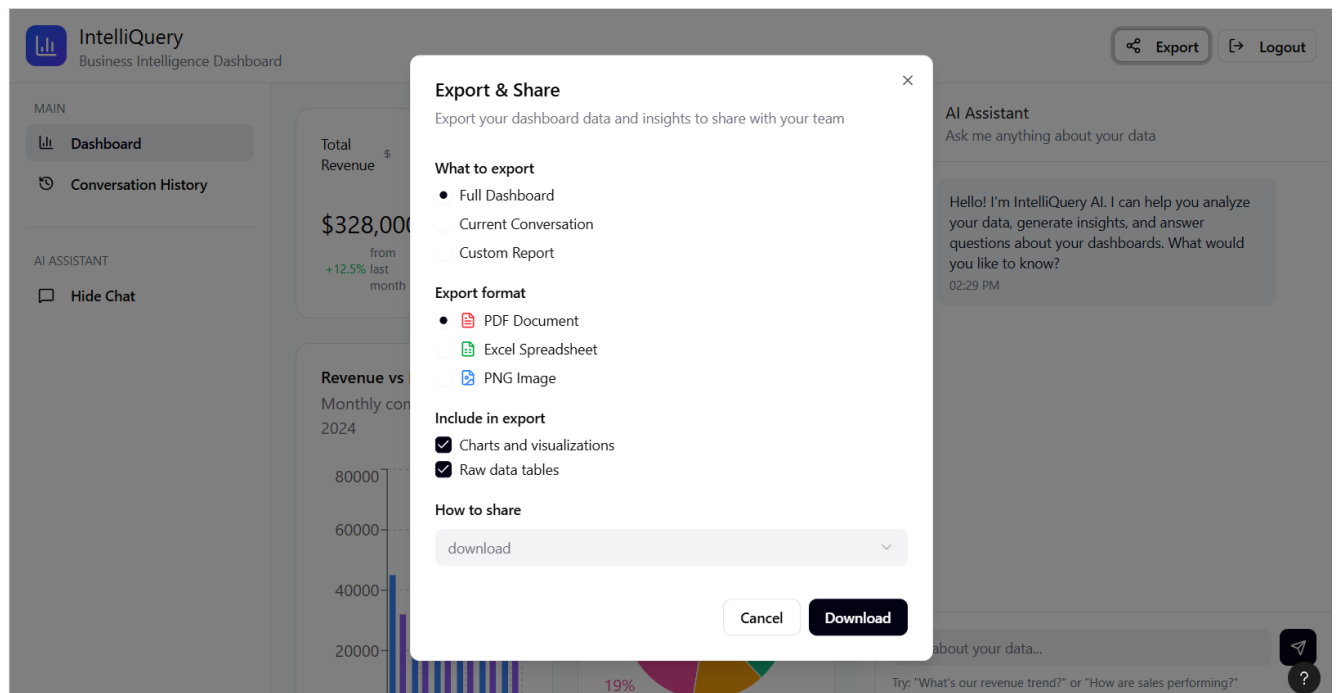


Figure 3.3: Export/Share dialog mockup

3.2 Administrator Portal

3.2.1 Dashboard (Home Page)

The administrator dashboard provides an overview of system usage, performance, and configuration status.

Components:

- **System Status:** Real-time indicators showing system health, API status, and resource utilization
- **Usage Analytics:** Charts showing user activity, query volume, and peak usage times
- **Active Users:** List of currently active users and their session information
- **Recent Alerts:** Notifications about system events, errors, or configuration changes

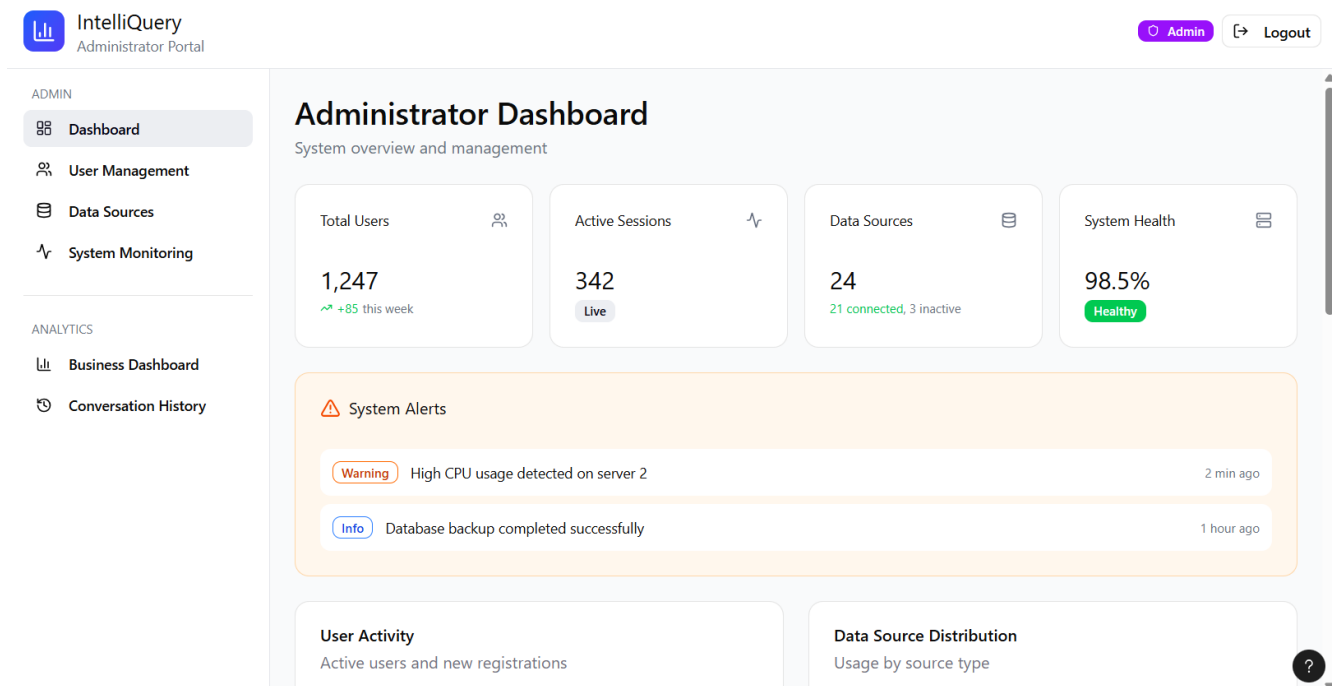


Figure 3.4: Administrator Dashboard mockup

3.2.2 User Management Page

- **User List:** Table displaying all users with columns for:
 - * Name and email
 - * Role assignment

- * Status (active/inactive)
- * Actions (edit, deactivate, reset password)
- **Add User:** Form for creating new user accounts with role selection
- **Bulk Operations:** Import users from CSV, bulk role assignments, bulk activation/deactivation
- **Search and Filter:** Search by name/email, filter by role or status

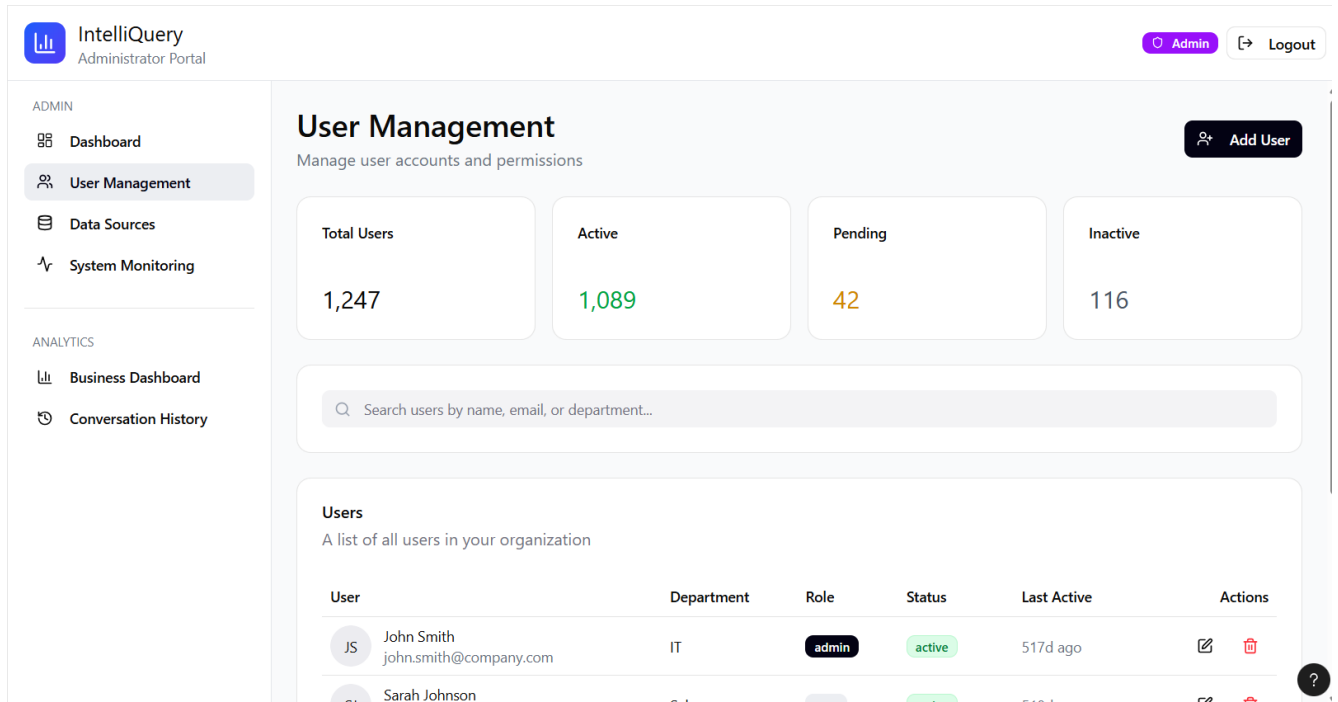


Figure 3.5: User Management page mockup

3.2.3 Data Source Configuration Page

- **Connected Data Sources:** Cards showing each configured Power BI workspace/dataset:
 - * Workspace name and dataset name
 - * Connection status (active/error)
 - * Last sync time
 - * Number of users with access
 - * Actions (edit, refresh, remove)
- **Add Data Source:** Wizard for connecting new Power BI workspaces:
 - * Azure AD authentication
 - * Workspace selection
 - * Dataset selection

- * Permission mapping
- * Connection testing
- **Permission Management:** Interface for mapping Power BI data permissions to IntelliQuery users

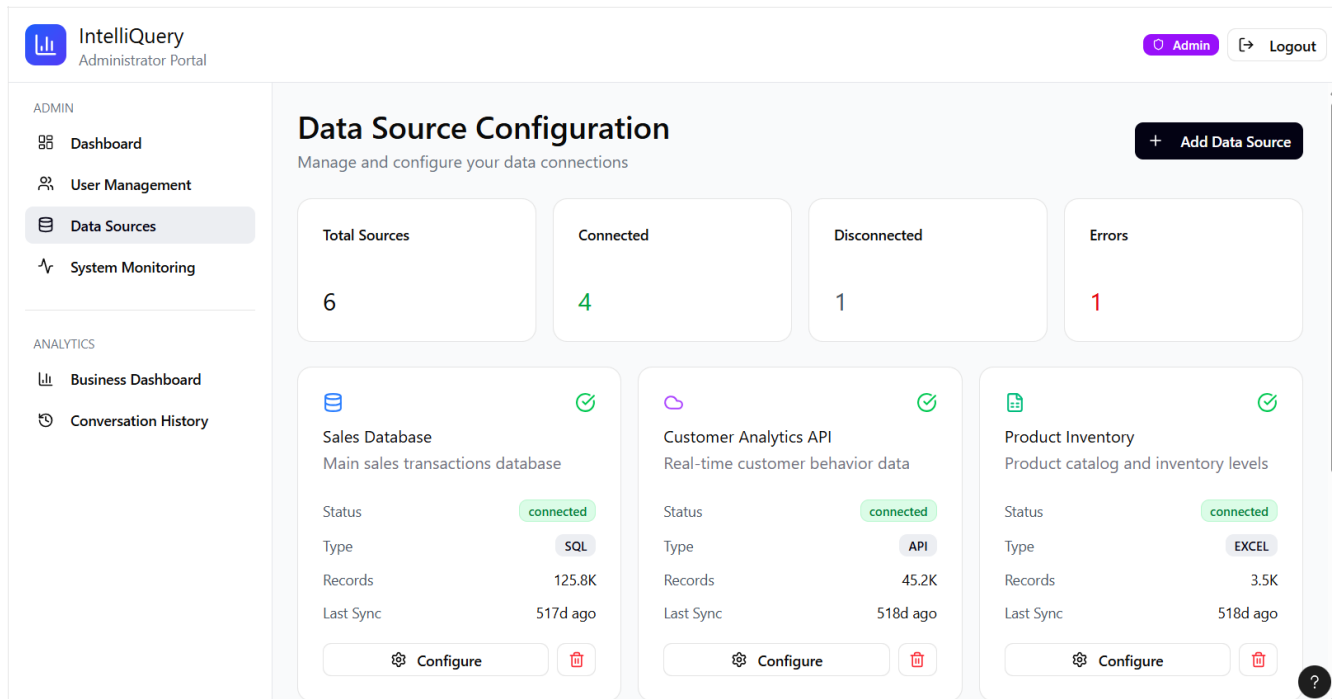


Figure 3.6: Data Source Configuration mockup

Chapter 4

Functional Requirements Specification

4.1 Stakeholders

- **Business Users:** Employees across all organizational levels who need access to business intelligence insights but lack technical expertise in SQL, DAX, or BI tools. This includes executives, managers, analysts, and operational staff.
- **Folio3 Technical Team:** Industry partners providing technical mentorship and validation of enterprise requirements.
- **Development Team:** Students and developers building and maintaining the Intelli-Query platform.
- **Academic Supervisors:** Faculty members providing guidance and ensuring academic rigor in project execution.

4.2 Actors and Goals

Actor	Goals
Business User	– Query organizational data using natural language – Receive accurate, relevant insights quickly – Conduct multi-step analysis through conversation – Export and share findings with colleagues – Track query history for reference

Actor	Goals
Administrator	– Manage user accounts and permissions – Configure connections to Power BI data sources – Monitor system performance and usage – Ensure data security and compliance – Troubleshoot issues and maintain system health
System	– Accurately interpret natural language queries – Generate appropriate visualizations – Maintain conversation context – Ensure data security and privacy – Provide reliable, performant service – Log activities for audit purposes

Table 4.1: Actors and Their Goals

Chapter 5

Use Cases

5.1 Business User Use Cases

5.1.1 Use Case 1: User Registration and Account Setup

Use Case ID	UC-BU-001
Use Case Name	User Registration and Account Setup
Goal	Allow new business users to create an account and complete initial setup for accessing IntelliQuery.
Actors	Business User, System
Preconditions	– User has been invited by an administrator – User has received invitation email with setup link – System is operational and accessible
Main Flow	1. User clicks on invitation link from email 2. System displays registration page with pre-filled email 3. User enters personal information (name, password) 4. User reviews and accepts terms of service and privacy policy 5. System validates input and creates user account 6. User is redirected to login page 7. System displays success message

Alternative Flows	2a. Expired Invitation Link – System detects expired invitation – System displays error message – System provides option to request new invitation – Administrator receives notification to send new invitation 4a. Weak Password – System detects password does not meet security requirements – System displays specific password requirements – User enters stronger password – Flow continues from step 5
Postconditions	– User account is created with appropriate role assignment – User can log in to system – User's profile is initialized with default settings – Administrator is notified of successful registration

Table 5.1: Use Case: User Registration and Account Setup

5.1.2 Use Case 2: Submit Natural Language Query

Use Case ID	UC-BU-002
Use Case Name	Submit Natural Language Query
Goal	Allow business users to ask questions about their data using natural language and receive relevant insights.
Actors	Business User
Preconditions	– User is logged in to the system – User has access to dataset

Main Flow	1. User types natural language query in chat interface 2. User submits query 3. System displays typing indicator 4. NLP Agent processes query 5. System validates permissions 6. System generates DAX query 7. Executes query 8. Visualization Agent selects chart 9. System displays visualization 10. System provides explanation 11. System suggests follow-up questions
Alternative Flows	4a. Ambiguous Query – System detects ambiguity – System asks for clarification – User selects intended meaning 5a. Insufficient Permissions – Permission error displayed – Suggest contacting admin – Log unauthorized attempt 7a. Query Execution Error – Power BI returns error – Log details – Display user-friendly message
Postconditions	– Query stored in history – Context updated – Metrics logged – Visualization available for export

Table 5.2: Use Case: Submit Natural Language Query

5.1.3 Use Case 3: Ask Follow-up Question

Use Case ID	UC-BU-003
Use Case Name	Ask Follow-up Question
Goal	Allow users to ask follow-up questions referencing previous context.
Actors	Business User
Preconditions	– User has an active conversation – Previous query exists – Context is available
Main Flow	1. User asks follow-up question 2. System retrieves context 3. Context Manager resolves references 4. NLP interprets question with context 5. System processes query 6. Results returned 7. Visualization updated
Alternative Flows	4a. Ambiguous Reference – System asks user to clarify reference – User clarifies
Postconditions	– Context updated – Follow-up stored in history

Table 5.3: Use Case: Ask Follow-up Question

5.1.4 Use Case 4: Export Insights

Use Case ID	UC-BU-004
Use Case Name	Export Insights
Goal	Allow users to export visualizations and insights.
Actors	Business User

Preconditions	– A visualization exists – User is viewing insights
Main Flow	1. User clicks export 2. System shows options 3. User selects format 4. Optional customization 5. User confirms export 6. System generates file 7. File downloads 8. Success message shown
Alternative Flows	6a. Export Error – System shows error – Retry offered
Postconditions	– File available – Activity logged

Table 5.4: Use Case: Export Insights

5.1.5 Use Case 5: View Conversation History

Use Case ID	UC-BU-005
Use Case Name	View Conversation History
Goal	Allow users to review their previous conversations.
Actors	Business User
Preconditions	– User is logged in – History exists

Main Flow	1. User opens history 2. System retrieves data 3. Conversations displayed 4. User filters/searches 5. User opens conversation 6. System loads it fully 7. User resumes or exports
Alternative Flows	2a. No History – Empty state shown – Suggest starting new conversation
Postconditions	– User accesses historical insights

Table 5.5: Use Case: View Conversation History

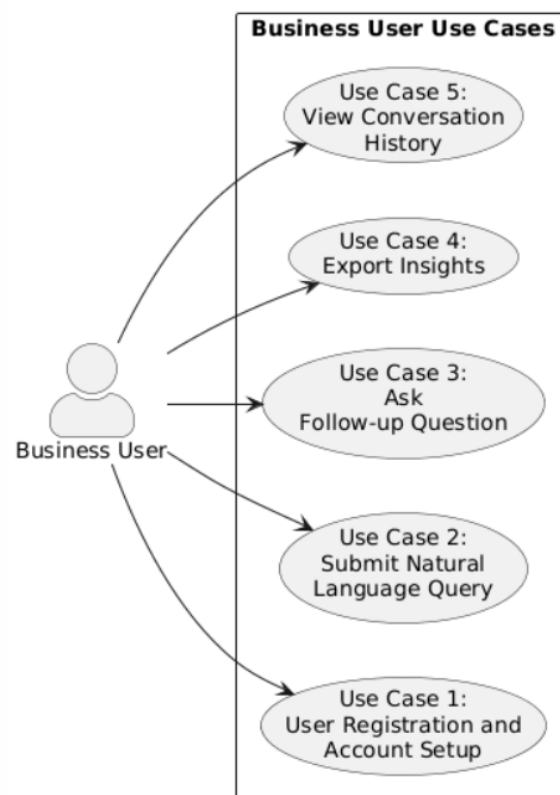


Figure 5.1: Use Case Diagram for Business User showing all BU use cases

5.2 Administrator Use Cases

5.2.1 Use Case 6: Administrator Login and Dashboard Access

Use Case ID	UC-ADM-001
Use Case Name	Administrator Login and Dashboard Access
Goal	Allow administrators to securely log in.
Actors	Administrator, Authentication Service
Preconditions	– Admin has valid credentials
Main Flow	1. Navigate to login 2. Enter credentials 3. System validates 4. Role verified 5. Session created 6. Redirect to dashboard 7. Dashboard displays metrics
Alternative Flows	3a. Invalid Credentials – Error shown – Retry with rate limiting 4a. Insufficient Privileges – Access denied – Attempt logged
Postconditions	– Admin logged in – Permissions active

Table 5.6: Use Case: Administrator Login and Dashboard Access

5.2.2 Use Case 7: Manage User Accounts

Use Case ID	UC-ADM-002
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Use Case Name	Manage User Accounts
Goal	Allow administrators to manage users.
Actors	Administrator, User Management Service
Preconditions	– Admin logged in – Has permissions
Main Flow	1. Open user management 2. View all users 3. Select action 4. Enter or modify user details 5. Validate 6. Apply changes 7. Notify user if needed
Alternative Flows	7a. Validation Error – Show specific error – Admin corrects 9a. Email Failure – Email failure logged – Warning shown
Postconditions	– User account updated – Audit trail updated

Table 5.7: Use Case: Manage User Accounts

5.2.3 Use Case 8: Configure Power BI Data Source

Use Case ID	UC-ADM-003
Use Case Name	Configure Power BI Data Source
Goal	Allow administrator to link Power BI workspace.
Actors	Administrator
Preconditions	– Admin logged in

Main Flow	1. Navigate to configuration 2. Add new data source 3. Azure authentication 4. Fetch workspaces 5. Select workspace 6. Fetch datasets 7. Select datasets 8. Configure permissions 9. Test connection 10. Save 11. Success message
Alternative Flows	4a. Auth Failure – Error shown – Retry 10a. Connection Fails – Error shown – Retry with adjustments
Postconditions	– Data source active – Users can query datasets

Table 5.8: Use Case: Configure Power BI Data Source

5.2.4 Use Case 9: View Audit Logs

Use Case ID	UC-ADM-004
Use Case Name	View Audit Logs
Goal	Allow administrators to review audit logs.
Actors	Administrator
Preconditions	– Admin logged in – Audit logging enabled

Main Flow	1. Open audit logs 2. View events 3. Filter logs 4. Search by keywords 5. Open detailed log entry 6. Export logs
Alternative Flows	None
Postconditions	– Admin reviews security events

Table 5.9: Use Case: View Audit Logs

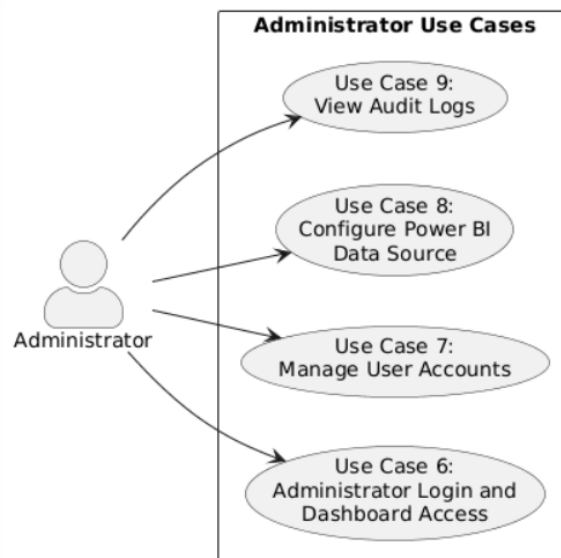


Figure 5.2: Use Case Diagram for Administrator

Chapter 6

User Interface Specification

6.1 Business User Interface Mockups

6.1.1 Login Screen

The login screen provides a clean, modern interface for user authentication.

Components:

- IntelliQuery logo and branding
- Email/username input field
- Password input field with show/hide toggle
- "Sign In" button
- "Forgot Password?" link

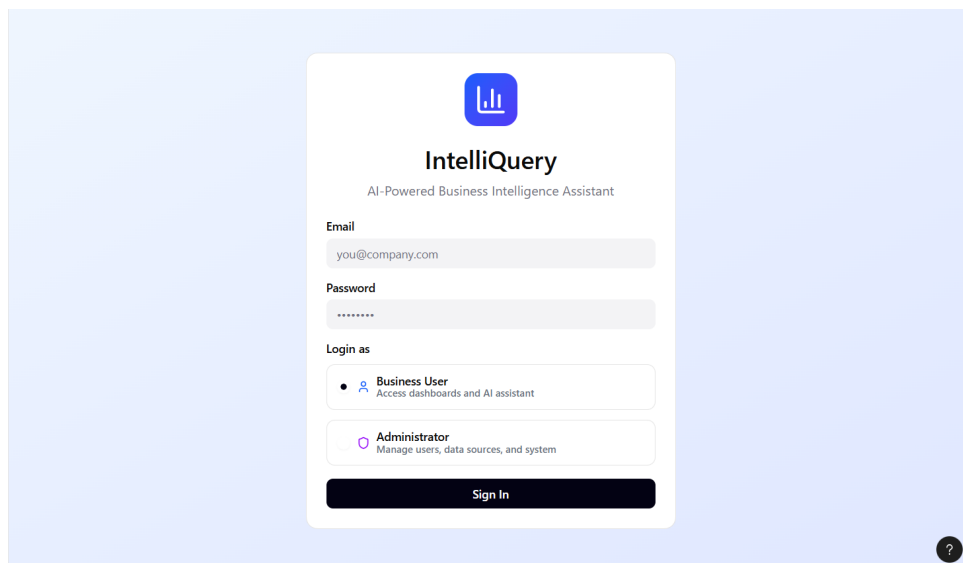


Figure 6.1: Login screen mockup showing clean, professional design

6.1.2 Business User Dashboard

The main dashboard is split into two primary panels for an optimal conversational BI experience, as already discussed in Section 3.1.1

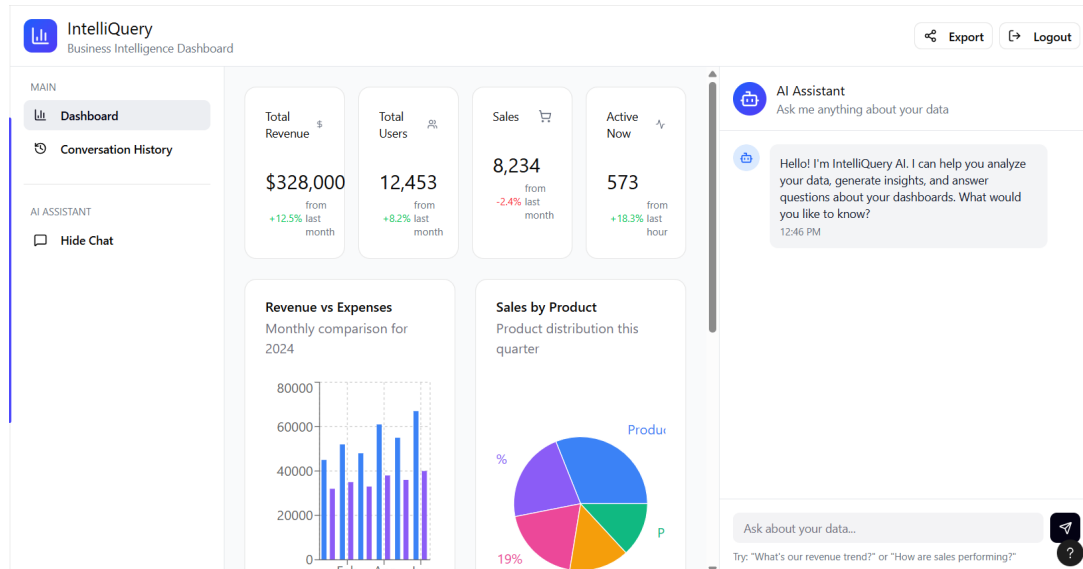


Figure 6.2: Main dashboard mockup with split-screen layout

6.1.3 Conversation History Page

Displays all previous conversations in a searchable, filterable grid as already discussed in Section 3.1.2

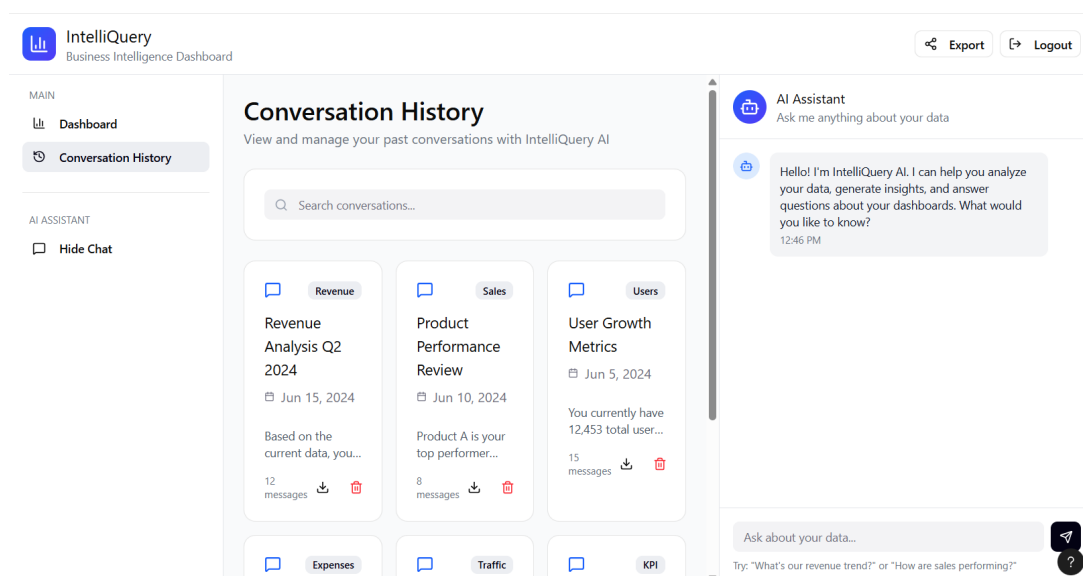


Figure 6.3: Conversation history page mockup with card layout

6.1.4 Export Dialog

Modal dialog for exporting insights with customization options as already discussed in Section 3.1.3

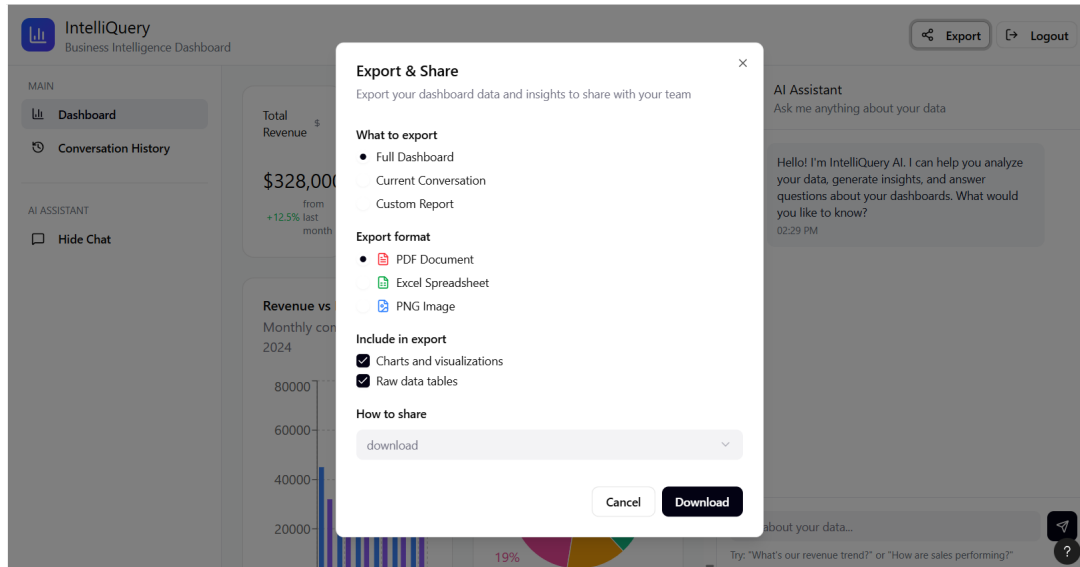


Figure 6.4: Export/Share dialog mockup

6.2 Administrator Interface Mockups

6.2.1 Administrator Dashboard

Provides comprehensive overview of system status and activity as already discussed in Section 3.2.1

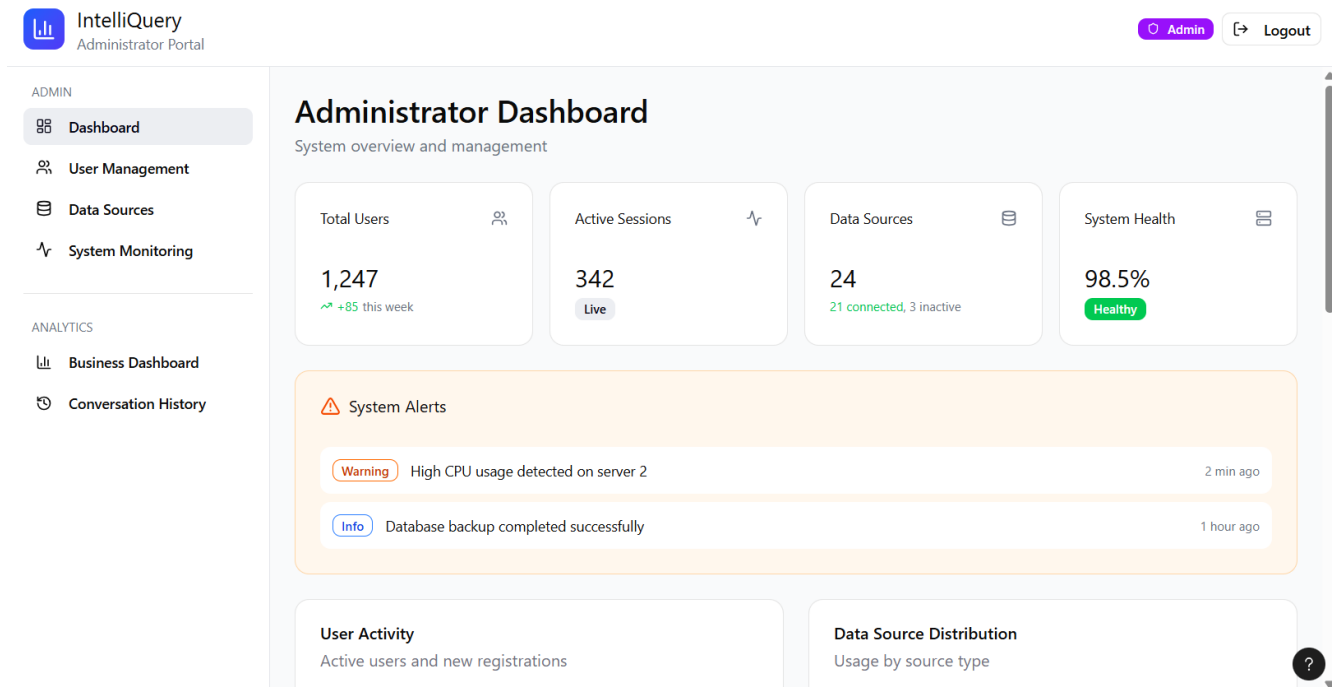


Figure 6.5: Administrator dashboard mockup

6.2.2 User Management Page

Interface for managing all user accounts as already discussed in Section 3.2.2

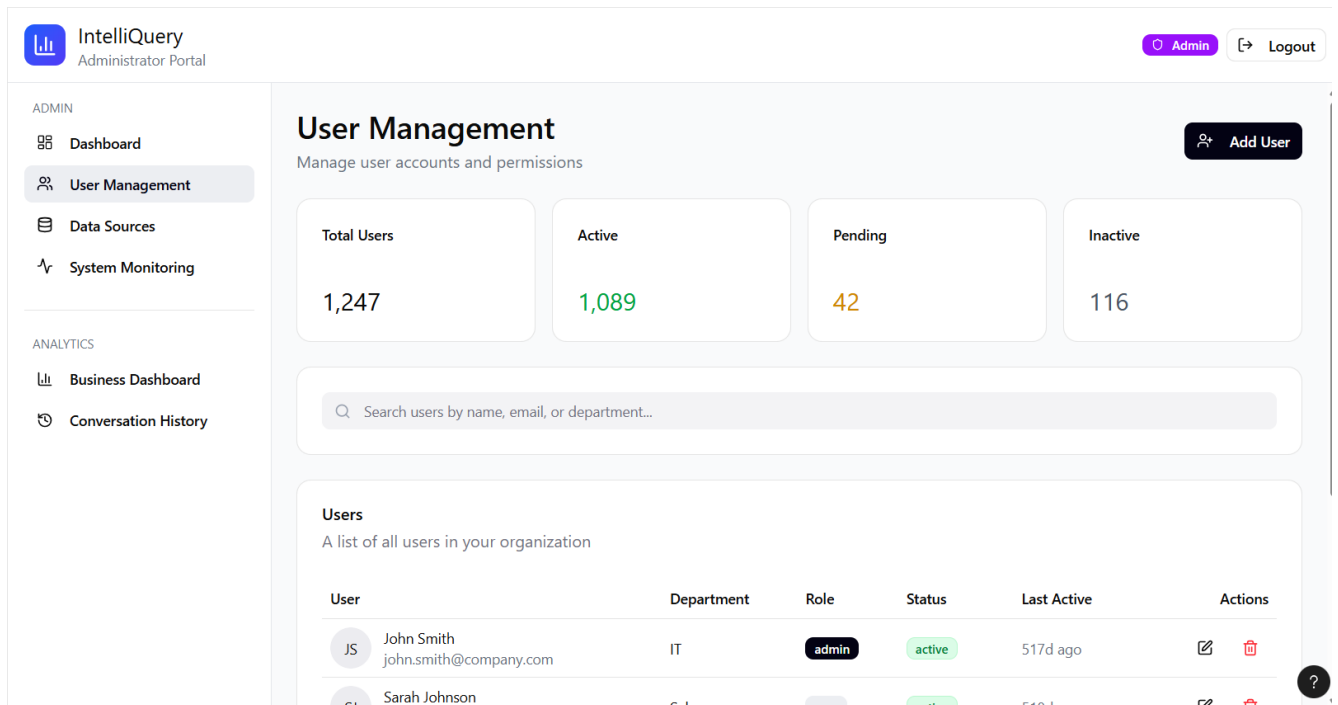


Figure 6.6: User management page mockup

6.2.3 Data Source Configuration Page

Interface for managing Power BI connections as already discussed in Section 3.2.3

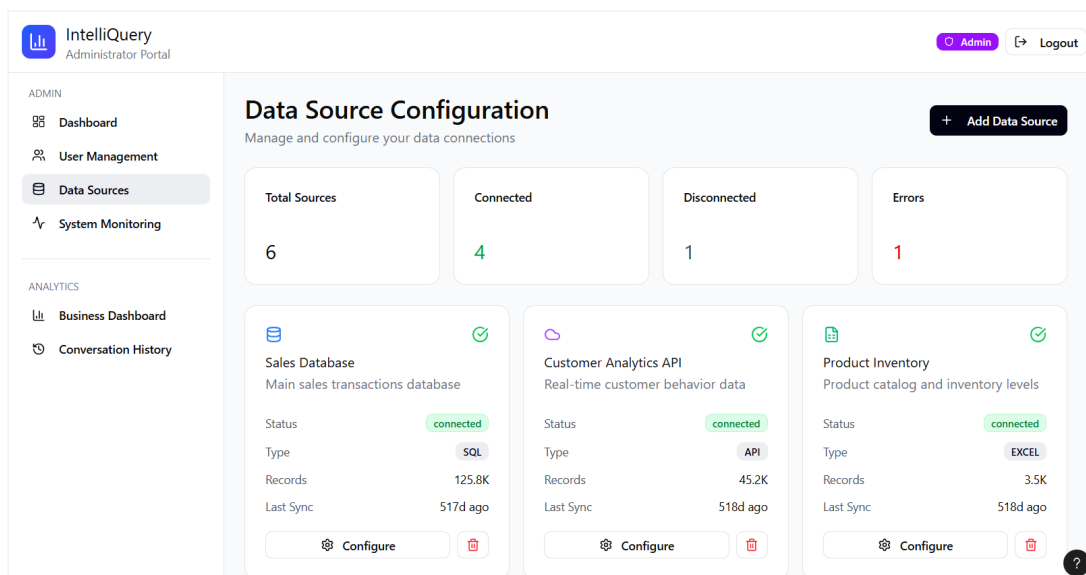


Figure 6.7: Data Source configuration page mockup

Chapter 7

Domain Analysis

7.1 Introduction to Conversational Business Intelligence

Business intelligence has traditionally been the domain of technical specialists who understand data modeling, query languages, and visualization techniques. The emergence of natural language processing and large language models presents an opportunity to democratize access to business intelligence by allowing users to interact with data using conversational natural language instead of complex technical interfaces.

Conversational BI represents a paradigm shift from static, pre-built dashboards to dynamic, interactive data exploration guided by natural dialogue. This approach aligns more closely with how business users naturally think about their data—through questions and conversations—rather than through technical constructs like tables, joins, and aggregations.

7.2 Challenges in Natural Language Query Understanding

Converting natural language to structured database queries presents several significant challenges that IntelliQuery must address:

7.2.1 Ambiguity Resolution

Natural language is inherently ambiguous. A question like "Show me sales for last quarter" could refer to different fiscal quarters depending on organizational context, could mean different sales metrics (revenue, units, profit), and could aggregate at different

levels (total, by product, by region). The system must either resolve these ambiguities through context or request clarification from users.

7.2.2 Entity Recognition and Mapping

Business users refer to data using business terminology that may not directly map to technical database schemas. "Revenue" might be stored as multiple calculated measures in Power BI; "last quarter" requires understanding the current date and organizational fiscal calendar; "top products" requires understanding ranking logic and thresholds. The system must maintain robust mappings between business terminology and technical implementations.

7.2.3 Complex Query Construction

Business questions often require complex queries involving multiple tables, aggregations, filters, and calculations. Converting "Show me year-over-year growth in revenue by region for products launched in the last 12 months" into a valid DAX query requires understanding temporal calculations, joins, filters, and aggregations—all inferred from natural language.

7.3 Context Management in Conversational Analytics

Unlike traditional BI interfaces where each report or dashboard is independent, conversational BI requires maintaining context across multiple interactions:

7.3.1 Short-term Context

Within a single conversation session, the system must remember what data the user is currently exploring, what filters have been applied, what metrics are being discussed, and what visualizations have been generated. This enables users to ask follow-up questions like "What about last year?" without repeating all context.

7.3.2 Long-term Context

Across multiple sessions, the system can learn user preferences, common query patterns, and frequently accessed data. This enables personalization and improved suggestion systems over time.

7.3.3 Reference Resolution

Users naturally use pronouns and references in conversation: "Show me that by region", "Compare these two", "Break it down further". The system must resolve these references to specific data elements, metrics, or visualizations from the conversation context.

7.4 Enterprise Integration Considerations

Integrating with existing enterprise BI infrastructure presents unique challenges:

7.4.1 Security and Permissions

IntelliQuery must respect existing Power BI security models, ensuring users can only access data they're authorized to see. This includes row-level security, column-level security, and object-level permissions. The system must validate permissions before executing queries and handle permission errors gracefully.

7.4.2 Performance and Scalability

Power BI APIs have rate limits and performance characteristics that must be considered. The system should implement intelligent caching, query optimization, and load balancing to ensure responsive performance even with multiple concurrent users.

7.4.3 Data Source Variability

Power BI datasets vary significantly in structure, naming conventions, and complexity. The system must be flexible enough to work with diverse data models without requiring extensive customization for each dataset.

7.5 Ethical Considerations in Conversational BI

Deploying AI-powered business intelligence systems raises several ethical considerations:

7.5.1 Transparency

Users must understand when they're interacting with AI-generated insights versus direct data retrieval. The system should provide confidence scores and explanations for its interpretations.

7.5.2 Bias and Fairness

LLMs can exhibit biases in interpretation. The system must be tested for fair interpretation across different business domains, terminology, and user groups.

7.5.3 Data Privacy

Conversation logs contain sensitive business information and must be protected with appropriate encryption, access controls, and retention policies.

7.6 Relevance to IntelliQuery

IntelliQuery's design directly addresses these domain challenges through:

- Advanced NLP pipeline with clarification dialogues for ambiguity
- Hierarchical context management for both short-term and long-term memory
- Intelligent visualization recommendation based on data and intent
- Deep Power BI integration with security model preservation
- Transparent AI interactions with confidence scoring
- Enterprise-grade security and privacy protections

Chapter 8

Technical Risk Scanning

8.1 Natural Language Processing Accuracy

8.1.1 Risk Description

The accuracy of intent classification and entity extraction directly impacts system usefulness. Misunderstanding user queries leads to incorrect results, user frustration, and potential business decisions based on wrong data.

8.1.2 Risk Factors

- Ambiguous or poorly structured user queries
- Domain-specific terminology not in training data
- Complex multi-part questions requiring decomposition
- Sarcasm, negation, or other linguistic nuances

8.1.3 Mitigation Strategies

- Fine-tune BERT models on business intelligence domain corpus
- Implement confidence scoring and request clarification when confidence is low
- Maintain feedback loop where users can correct misinterpretations
- Build comprehensive test suite covering edge cases and complex queries
- Implement fallback to simpler interpretation strategies when advanced NLP fails

8.2 Power BI API Dependencies

8.2.1 Risk Description

System functionality depends entirely on Power BI REST API availability and performance. API changes, rate limiting, or outages directly impact IntelliQuery's ability to serve users.

8.2.2 Risk Factors

- Power BI API rate limits during peak usage
- API version deprecation requiring system updates
- Authentication token expiration causing query failures
- Network latency between IntelliQuery and Power BI services

8.2.3 Mitigation Strategies

- Implement robust caching strategy to reduce API call frequency
- Design retry logic with exponential backoff for transient failures
- Monitor API changes and maintain version compatibility layer
- Implement queue-based architecture to manage API rate limits
- Provide graceful degradation when Power BI is unavailable
- Maintain comprehensive error logging for troubleshooting API issues

8.3 Academic Timeline Constraints

8.3.1 Risk Description

Project must be completed within 22-week academic timeline. Scope too ambitious for timeline results in incomplete delivery.

8.3.2 Risk Factors

- Underestimating development complexity
- Scope creep adding features beyond MVP

- Integration challenges taking longer than expected
- Team coordination and resource constraints

8.3.3 Mitigation Strategies

- Clear MVP definition focusing on core functionality
- Agile development with 2-week sprints
- Regular progress reviews with supervisors
- Risk buffer time in schedule
- Parallel development where possible
- Clear division of responsibilities among team members
- Prioritized feature backlog with "must-have" vs. "nice-to-have"
- Plan for scaled-back version if timeline pressures increase

Chapter 9

Ethical Considerations

9.1 Data Privacy

Measures: TLS 1.3 encryption, AES-256 at rest, RBAC, data anonymization for LLM queries, GDPR compliance

9.2 Transparency

Measures: Display confidence scores, explain query interpretations, show generated DAX queries, provide audit trails

9.3 Bias and Fairness

Measures: Test across diverse business domains, monitor success rates by user group, implement feedback mechanisms

9.4 Accountability

Approach: Position as decision support tool (not decision-maker), maintain audit trails, enable human override of AI interpretations

Chapter 10

Competitive Analysis

Feature	Power BI Copilot	ThoughtSpot	IntelliQuery
Context Memory	Basic	Limited	Advanced hierarchical
Custom Dashboard	No	No	Yes, purpose-built
Conversation Flow	Single-turn	Search-based	Multi-turn dialogue
Cost	Premium	High	Enterprise-friendly
Infrastructure	Native	Proprietary	Uses existing Power BI

Table 10.1: Competitive Comparison

10.1 IntelliQuery Differentiation

1. **Advanced Context:** Hierarchical memory for coherent multi-turn conversations
2. **Custom Interface:** Built for conversation, not adapted from dashboards
3. **Agentic Architecture:** Specialized AI agents for flexibility and maintainability
4. **Cost-Effective:** Works with existing Power BI without proprietary infrastructure
5. **Industry Partnership:** Real-world validation through Folio3 collaboration

10.2 Key Design Considerations

10.2.1 Accuracy and Reliability

High accuracy in query interpretation is non-negotiable. The system must provide reliable, accurate insights to build user trust and enable confident decision-making.

Implementation Approaches:

- Fine-tuned BERT models for business domain
- GPT-4 for complex query understanding
- Multi-stage validation pipeline
- Confidence scoring for all interpretations
- User feedback loops for continuous improvement

10.2.2 User Experience

The interface must be intuitive enough that non-technical users can be productive with minimal training while remaining powerful enough for complex analytical workflows.

Implementation Approaches:

- Clean, uncluttered interface design
- Context-aware query suggestions
- Progressive disclosure of advanced features
- Consistent visual language throughout
- Comprehensive but optional help system

10.2.3 Performance

System must respond quickly to maintain conversational flow. Users will abandon slow systems regardless of accuracy.

Implementation Approaches:

- Multi-level caching strategy
- Asynchronous processing where appropriate
- Optimized database queries
- Efficient algorithms for context management
- Load balancing and horizontal scaling capabilities

10.2.4 Scalability

Architecture must support growth from pilot deployment to enterprise-wide adoption.

Implementation Approaches:

- Microservices architecture
- Containerization with Docker
- Stateless design where possible
- Database optimization and indexing
- Cloud-native deployment strategies

10.3 Addressing Project Constraints

10.3.1 Academic Timeline

With 22 weeks for development, we must prioritize ruthlessly:

MVP Features (Must Have):

- Basic natural language query processing
- Simple context management within single conversation
- Connection to one Power BI dataset
- Basic visualizations (bar, line, table)
- User authentication
- Administrator console for basic configuration

Enhanced Features (Should Have):

- Advanced context management across sessions
- Multiple visualization types with recommendations
- Multiple dataset support
- Export functionality
- Conversation history

Advanced Features (Nice to Have):

- Voice input
- Advanced analytics (trends, predictions)
- Collaborative features (sharing, commenting)
- Mobile app

10.3.2 Resource Constraints

Limited budget requires creative solutions:

Strategies:

- Use academic licenses where available
- Leverage free tiers of cloud services during development
- Implement aggressive caching to minimize API costs
- Consider hybrid approach with open-source models for some tasks

10.3.3 Technical Expertise

Team is learning advanced technologies simultaneously with development:

Strategies:

- Allocate learning time in schedule
- Focus each team member on specific technical areas
- Leverage Folio3 mentorship for guidance
- Use proven libraries and frameworks over custom solutions
- Maintain good documentation for knowledge sharing

Chapter 11

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